

How the World Cup 2026 Prediction Model Works

A complete guide to the mathematical methods behind the joint score probability engine

WizardOfOdds.com

June 2026

How the World Cup 2026 Prediction Model Works

Predicting the outcome of a football match is an exercise in structured humility. A single match produces roughly two or three goals. Those goals arrive from a chaotic sequence of decisions, deflections, and individual moments of brilliance that no model can fully capture. Even the best football models – the ones used by sharp sportsbooks and professional betting syndicates – are wrong more often than they are right about any individual game.

What a good model can do is get the probabilities right over a large number of matches. It can assign 30% probability to outcomes that happen 30% of the time, and 70% probability to outcomes that happen 70% of the time. It can be faster than the public market in incorporating new information, and it can identify specific markets where the bookmaker's price does not reflect the available evidence. That is a narrower ambition than predicting winners – but it is the ambition that leads to long-run results.

This model produces a joint probability distribution over every possible final score for every 2026 FIFA World Cup match. It does this by running six competing parametric score models daily, blending them with a composite rating system built from six independent data sources, reconciling the result against live bookmaker prices, and running every output through an automated calibration framework. The pipeline runs without human intervention 24 hours a day.

The following sections describe every stage of that process, from the first rating inputs to the final edge report you see on the site.

The Output: What a Joint Score PMF Is

Before describing how the model works, it helps to understand precisely what it produces.

A PMF is a Probability Mass Function – a list of all possible outcomes with a probability assigned to each one. For a football match, the joint score PMF is a grid. Each cell (h, a) in that grid holds the probability $P(\text{Home goals} = h, \text{Away goals} = a)$ – the probability that the home team scores exactly h goals and the away team scores exactly a goals in regulation time.

The grid runs from 0 to 15 goals per team, plus an explicit tail-mass bucket for extreme scores beyond that range. Every cell, including the tail, must sum to exactly 1.0 – a constraint enforced at every stage of the pipeline.

Every betting market on the site flows mechanically from this grid. Over 2.5 goals? Sum all cells where $h + a \geq 3$. Both Teams to Score? Sum all cells where $h \geq 1$ and $a \geq 1$. Home Win? Sum all cells where $h > a$. A correct score bet on 2-1? That is a single cell: $P(h = 2, a = 1)$. There is no separate model for each market type. The PMF is the entire model, and every market is a different question about the same distribution.

This architecture matters. It means all the markets are internally consistent by construction. The model cannot simultaneously estimate Home Win at 45% and correct-score probabilities that sum to 52% across all home-win scorelines – an inconsistency that appears frequently on sites that blend data from different sources.

Step 1: Rating Every Team – The Composite Prior

The first task is assigning each of the 48 World Cup teams an attack rating and a defense rating. Specifically, the model estimates λ_{att} (expected goals scored against an average opponent on a neutral pitch) and λ_{def} (expected goals conceded). The World Cup-calibrated global average is 1.30 goals per team per match, derived from the 128 matches played across the 2018 and 2022 tournaments.

No single rating system is good enough to trust entirely. Club data does not transfer cleanly to international football. FIFA rankings can lag months behind a team's recent form. Bookmaker odds contain real information but can be distorted by public betting bias or late market movements. The solution is to build a composite prior from six independent sources and let their collective weight produce a more stable estimate than any one of them could alone.

Source 1: Market-Implied Strength (20% weight)

When bookmaker odds are available for a match, they contain information that no rating system has: late team news, undisclosed injuries, sharp money, and the collective opinion of everyone in the world who has taken a view on the match. The model extracts market-implied attack and defense lambdas by reverse-engineering what team strengths would produce the bookmaker's 1X2 probabilities.

Before doing this, the bookmaker margin must be removed. Raw quoted odds always include a margin of roughly 5-8% that makes them look less certain than they actually are. SHIN normalization is applied to all odds from all bookmakers to convert them to fair (no-margin) probabilities. This is not optional – using raw odds as fair probabilities would consistently bias every market comparison.

Up to 6 sportsbooks contribute to each match's market picture: FanDuel, DraftKings, BetMGM, BetRivers, Caesars, and Fanatics. Their no-vig probabilities are averaged to produce a consensus view.

The 20% weight for market-implied strength is empirically derived, not arbitrary. Analysis of completed matches showed that using zero market weight produced a mean absolute difference of 18.3 percentage points between the model's PMF-implied win probabilities and the market's no-vig 1X2. Raising the market weight to 20% cut that gap to 15.1 percentage points while producing better closing line value positioning than higher weights. More market weight did not help – it started pulling the model too far toward the market and reducing its independent predictive contribution.

Source 2: FIFA World Ranking – March 2026 (roughly 12% weight)

FIFA's official points system, converted to an attack lambda through a calibrated sigmoid mapping. The FIFA ranking captures long-run international performance across all competitions and provides a useful signal particularly for teams with limited recent competitive data. Its weakness is that rankings update infrequently and can reflect performance from years ago. The model uses the March 2026 snapshot – the last official update before the tournament.

Source 3: Qualifying Record (roughly 10% weight)

Each team's attack and defense efficiency during their qualifying campaign, shrunk toward their confederation's average using Bayesian shrinkage. The degree of shrinkage scales with the number of competitive matches played – a team with 18 qualifying matches gets treated very differently from a team with 6. A nation that dominated its qualifying group is genuinely more dangerous than one that scraped through on goal difference, and this source captures that.

Source 4: Pi Rating (roughly 30% of remaining weight)

The Pi rating, computed using the `penaltyblog` library, is a goal-margin-sensitive variant of Elo. Unlike standard Elo, which updates based only on match result (win, draw, loss), Pi rating updates based on the actual score. A team that wins 4-0 earns a larger rating boost than one that wins 1-0. This makes Pi more responsive to genuine performance shifts and better suited for a model whose primary target is predicting scores rather than just results.

Source 5: Elo Rating (roughly 55% of remaining weight after Pi)

Classic international Elo, also from `penaltyblog`, based on win/loss/draw outcomes only. Elo is more stable over long periods than Pi precisely because it does not react to score margins – a team cannot manipulate its Elo by running up the score. It serves as a stabilizing floor that prevents the prior from swinging too sharply on an unusual result, and it is the dominant signal for teams whose Pi and market data are sparse.

Source 6: Confederation Baseline (roughly 5% as a soft floor)

Historical World Cup attack averages by confederation provide the final input: CONMEBOL 1.45 goals per team per match, UEFA 1.35, CONCACAF 1.20, CAF 1.10, AFC 1.10, OFC 0.90. Defense baselines mirror these figures. This source acts as a soft floor – it prevents any team from receiving a lambda estimate that is wildly inconsistent with what teams from its region have historically produced at World Cups. For well-rated teams with extensive data, this source contributes almost nothing. For teams where other sources are sparse, it prevents unrealistic extreme estimates.

When market odds are unavailable for a match, the market-implied source is excluded and the remaining five sources expand proportionally to fill its 20% share.

Host Advantage and Venue Adjustments

USA, Canada, and Mexico each receive a co-host adjustment of +0.10 to attack lambda and -0.10 to defense lambda. This reflects the measurable advantages of home geography – minimal travel,

familiar conditions, and genuine crowd support – without overstating an effect that is smaller in a multi-host tournament than in a traditional single-host World Cup.

Three Mexican venues sit at elevations that materially reduce scoring rates for unacclimated players. Both teams' lambdas are scaled equally at these venues, since neither side has an acclimatization advantage:

- Estadio Azteca in Mexico City sits at 2,230 meters, carrying a 0.93x multiplier – approximately a 7% scoring reduction.
- Estadio Akron in Guadalajara sits at 1,560 meters, carrying a 0.97x multiplier – approximately a 3% reduction.
- Estadio BBVA in Monterrey sits at 530 meters, which produces no adjustment.

Tournament Adjustment: Learning from 2026 Results

Once 2026 World Cup matches are completed, the model incorporates their results into each team's composite rating through a Bayesian shrinkage mechanism.

For a team that has played n World Cup matches in 2026, the tournament adjustment factor is shrunk toward 1.0 (no adjustment) with weight $n / (n + 3)$. The number 3 is the prior strength – equivalent to saying “three prior-data matches are worth about as much as one actual tournament match.” A team that played 1 match gets weight $1 / (1 + 3) = 0.25$, meaning 75% of the raw adjustment is discarded in favor of the prior. A team that played 3 matches gets weight $3 / (3 + 3) = 0.50$. This prevents overreaction to a single game while genuinely updating on multiple games.

Adjustments are capped at +/-30% in either direction. As of the current date, 16 teams have active tournament adjustments based on 8 completed 2026 World Cup matches.

The 2026 matches themselves also receive substantially higher training weight than historical data. The Dixon-Coles time-decay parameter ξ is set to 0.010 for 2026 matches versus 0.0018 for historical matches – meaning each completed 2026 match is roughly 5 times more influential on the model's parametric structure than a match from 2018 or 2022.

Step 2: The Parametric Model Competition

With composite team ratings in hand, the model does not simply run a single Poisson calculation. It runs six competing parametric models – all fitted on the combined historical and 2026 training data – and selects the one that best explains the available World Cup evidence.

The six competitors are:

Independent Poisson is the classical baseline. Each team's goals are an independent Poisson random variable with mean equal to its estimated lambda. It is the simplest model, and a useful benchmark – if nothing else beats it on the data, Occam's Razor says use it.

Dixon-Coles extends the Poisson model with a correlation parameter ρ that adjusts the joint probability of low-scoring outcomes: 0-0, 1-0, 0-1, and 1-1. When ρ is negative, these scorelines are more probable than independent Poisson predicts. When ρ is zero, the model reduces to independent Poisson. The current calibrated ρ from World Cup data is 0.0, indicating that the

goal-independence assumption performs adequately in high-stakes international matches and no low-score correction is warranted.

Bivariate Poisson models positive correlation between home and away goals through a shared latent component. It captures the idea that certain match conditions simultaneously lift both teams' scoring rates.

Weibull Copula models each team's goals using a Weibull distribution rather than Poisson, then links them through a copula. The Weibull distribution can accommodate heavier tails than Poisson – more high-scoring outliers than the Poisson distribution would predict.

Negative Binomial relaxes the Poisson constraint that the variance equals the mean. Empirically, many football datasets show overdispersion – goal variance exceeding goal mean – which the Negative Binomial captures directly.

Zero-Inflated Poisson adds explicit probability mass at zero to account for scoreless outcomes occurring more frequently than strict Poisson predicts. This can matter for matches involving defensive teams at altitude, where the probability of 0-0 may genuinely exceed the Poisson estimate.

Each model is evaluated against held-out World Cup data using a walkforward backtest – predicting each match from what was known before it was played, then comparing to the actual result. The model with the lowest negative log-likelihood (the standard information-theoretic measure of how surprised a model was by what actually happened) is selected as the daily champion and used to generate all upcoming predictions.

The champion can change day to day. As more 2026 matches accumulate, the evaluation becomes more informative, and the model naturally specializes to the specific conditions of this tournament rather than being fixed to patterns from 2018 and 2022.

Step 3: Market Reconciliation

The composite prior produces strong fundamental ratings. The bookmaker market produces something different: real-time information that no rating system can access. A team announcement three hours before kickoff that the first-choice goalkeeper is injured, a tactical leak, sharp money moving from London to Hong Kong – all of this gets absorbed into market prices before anyone publishes a statistical update. Ignoring available market information is leaving real signal on the table.

Market reconciliation incorporates this information without simply replacing the model with the market. For each match, the model extracts fair market probabilities from all available bookmakers (after SHIN normalization) across as many market types as the API carries: 1X2 result, over/under totals from 0.5 through 6.5, both teams to score, draw no bet, double chance, and correct score lines.

A constrained optimization algorithm (SLSQP – Sequential Least Squares Programming) then adjusts the initial parametric PMF grid to satisfy these market constraints while minimizing the Kullback-Leibler divergence from the original distribution. KL divergence measures how far the adjusted distribution has moved from the starting point – minimizing it ensures the optimizer only moves as far from the parametric prior as the market evidence requires. The constraints are hard: all cell values must remain non-negative, and they must sum to 1.0.

The result is a distribution that respects both the model's structural estimates and the market's current pricing. When SLSQP converges to a better solution than a simple weighted-average blend of the two distributions, SLSQP is used; otherwise the blend is used. In practice, SLSQP dominates for matches where the market's implied probabilities differ materially from the parametric prior – precisely the matches where market information is most valuable.

Step 4: Calibration

A model is calibrated when the probabilities it outputs match observed frequencies. When it says 30%, outcomes should occur 30% of the time. When it says 70%, they should occur 70% of the time.

Calibration is distinct from accuracy. An overconfident model might assign 80% probability to outcomes that only occur 65% of the time – it gets the ranking right (80% outcomes happen more often than 60% ones) but the numbers are systematically too extreme. An underconfident model has the opposite problem.

The model uses temperature scaling to correct this. A single parameter T is fitted by minimizing exact-score log loss on out-of-sample predictions from the 2018 and 2022 World Cups. The calibrated probability for each cell in the grid is:

$p_{\text{calibrated}}(h, a) = p_{\text{raw}}(h, a)^{(1/T)}$, then renormalized to sum to 1.0.

When $T > 1$, the exponent $(1/T)$ is less than 1, which compresses the distribution – it flattens high probabilities down and raises low probabilities up, correcting for overconfidence. When $T < 1$, it sharpens the distribution, correcting for underconfidence. $T = 1$ leaves everything unchanged.

The current calibrated temperature is $T = 1.127$. The raw parametric output is mildly overconfident – it assigns somewhat too much probability to the most likely outcomes and somewhat too little to the less likely ones. This is a common pattern in football models trained on limited data; the parametric structure tends to be too certain. The temperature correction is modest, indicating that the raw model is reasonably well-calibrated and the adjustment is a fine-tuning rather than a major revision.

The Dixon-Coles ρ parameter is similarly calibrated from actual World Cup data rather than using a default value from league football. The calibrated ρ of 0.0 means no low-score correction is applied.

Calibration is evaluated on five metrics: exact-score log loss (the primary metric), Ranked Probability Score for the 1X2 market, multiclass Brier score, Expected Calibration Error, and the ignorance score. All are computed on out-of-sample predictions only.

One honest caveat: 128 World Cup matches is a small dataset. Calibration metrics computed from 128 observations carry meaningful statistical uncertainty. The temperature correction is a useful adjustment on average, but it is not a guarantee of statistical precision at any specific probability level. As the 2026 tournament adds completed matches, the calibration sample grows and these estimates become more reliable.

Step 5: Edge Screening

With a calibrated PMF for each match, the model compares its probability estimates against the bookmaker's no-vig prices. An edge exists when the model's estimate is higher than the market's:

$$\text{Edge} = (\text{Model probability} - \text{Market implied probability}) / \text{Market implied probability}$$

An edge of 0.08 means the model estimates the outcome is 8% more likely than the market is pricing, relative to the market price – not 8 percentage points more likely, but 8% proportionally higher. A market showing 40% implied probability with an 8% edge would have a model probability of 43.2%.

Raw edge alone is not enough to flag a market as a value opportunity. The model applies three simultaneous filters:

Edge $\geq$ 4% sets a minimum threshold for meaningful signal above the noise of estimation error. Markets below this level may reflect model uncertainty rather than genuine mispricing.

Confidence interval check requires that the lower bound of the model's 90% confidence interval still exceeds the market implied probability. The CI is computed by varying each team's lambda by $\pm 12\%$ – a conservative estimate of lambda uncertainty – and measuring the resulting range of PMF-derived probabilities. A market passes only when even the pessimistic scenario (lambdas shifted against the edge) still shows the model above the market price. This filter removes markets where the apparent edge is entirely inside the model's uncertainty range.

Market implied $> 2\%$ excludes very low-probability markets where thin liquidity and high variance make edge estimates unreliable.

Markets passing all three filters are flagged as value opportunities and sized using the Kelly criterion:

$$\text{Full Kelly fraction} = \text{Edge} / (\text{Decimal odds} - 1)$$

$$\text{Half Kelly} = \text{Full Kelly} / 2$$

All stakes capped at 5% of bankroll.

Half Kelly is the default because the $\pm 12\%$ lambda uncertainty is a fixed prior assumption, not an empirically measured per-match figure. The true lambda uncertainty for a given match may be higher or lower, but the model has no way to distinguish between them precisely. When uncertainty is higher than assumed, Full Kelly systematically overbets. The 5% hard cap provides a backstop regardless of what the formula computes.

Step 6: Closing Line Value – The Industry Standard for Evaluating Model Edge

Counting wins and losses is a poor way to evaluate a prediction model. A model that goes 60% on 1X2 bets might simply have been lucky to focus on heavy favorites. A model that goes 48% might have been consistently finding genuine value on underdogs and still be profitable in the long run.

The industry measure that strips luck away from skill is Closing Line Value, or CLV.

The closing line is the bookmaker's final odds immediately before kickoff – specifically, the no-vig (fair) probability after removing the bookmaker margin. This is the market's best collective estimate

of the true probability, having absorbed every piece of publicly available information. It represents the aggregated opinion of every sharp bettor, quantitative model, and market maker who has looked at the match.

CLV measures whether a model's predictions were better than the closing market. If the model assigns 55% probability to a match outcome and the closing no-vig price implies 45%, the model was predicting meaningfully higher probability than where the full-information market ultimately settled. That positive CLV is evidence of genuine edge – not because the model was necessarily right about the specific outcome, but because it was ahead of the market's information flow.

A model that consistently beats the closing line across a large sample of markets is not getting lucky. It is providing information that the market was slow to price in. Consistently losing to the closing line – predicting lower probability than where the market closes – means the market was consistently smarter than the model, and the model's positive edges were illusions.

For every scheduled match, the model records:

- Its predicted probabilities at the time of prediction (with opening odds from the market at that time)
- The closing odds captured approximately 3 minutes before kickoff via an automated pre-match pipeline

For each of 15 markets per match – the three 1X2 outcomes, BTTS Yes and No, Over/Under at 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, and 6.5 goals, and Under at 1.5, 2.5, and 3.5 – the model computes two numbers:

Model vs. closing is the model probability at prediction time minus the closing fair probability. A positive number means the model was predicting a higher probability than where the market ultimately settled.

Opening drift is the closing probability minus the opening probability. Large drift means significant information moved the market between the time the model made its prediction and kickoff. A model that consistently calls the direction of market drift – predicting high probabilities that the market later moves toward – is demonstrating forward-looking accuracy.

The model currently tracks 71 matches across all 15 markets per match, with 94% of records having opening market odds available.

The Automated Pipeline: A Living System

Every stage described above runs automatically, without human intervention, around the clock. The pipeline is not a tool that runs when someone presses a button – it is a continuously running system that treats each day of the World Cup as a new data point to incorporate.

Daily retraining at 8:00 AM UTC is the full rebuild. The pipeline fetches fresh data from the API: match results, bookmaker odds from up to 6 vendors, team statistics, expected goals, shot data, match events, and group standings. It retrains all six parametric models on the combined historical plus 2026 dataset with the latest completed matches included. It rebuilds the composite prior. It regenerates predictions for every upcoming fixture. It logs calibration metrics and deploys the updated prediction JSON files to the server. Yesterday's results are in today's predictions.

Hourly updates every hour of the day run a lighter refresh. These capture odds movements from bookmakers, update CLV tracking records, and rerun predictions if newly completed matches have appeared since the daily run. The hourly cycle means the site's market comparisons reflect the latest bookmaker pricing even when the daily retraining has already completed for the day.

Live snapshots every 2 minutes during match hours are how the live in-play page stays current. Match hours run from approximately 9:00 AM to 3:00 AM Eastern Time to cover the full range of World Cup kickoffs. During these hours, a live snapshot pipeline runs every 2 minutes. When a match is detected in progress, the pipeline self-chains – each completed run immediately triggers the next – ensuring near-continuous updates rather than depending entirely on the scheduled cron system, which can throttle during periods of high demand. Each run fetches the current score and match clock, computes the conditional PMF, and uploads the live JSON to the server.

Closing odds capture at T-3 minutes before each kickoff is a separate dedicated pipeline. It watches the match schedule every 15 minutes. When a match is detected within the next 15 minutes, it sleeps until exactly 3 minutes before kickoff, fetches the final market odds across all 15 tracked markets, applies SHIN normalization, records the closing probabilities, and commits the updated CLV records. It also bootstraps the live snapshot chain by triggering the first live run at kickoff, ensuring the live page is active from the opening whistle even if the regular cron was throttled.

Honest Limitations

No model of football should be presented without an honest accounting of what it cannot do.

All probabilities represent regulation time only. Extra time and penalty shootouts are not modeled. For knockout matches that go to extra time, the regulation-time distribution described here is not the full story of match outcomes.

The independence assumption is an approximation. The parametric structure treats home and away goals as generated by two separate, uncorrelated processes. In reality, a goal immediately changes both teams' incentives and triggers tactical adjustments. The live model's score-state multipliers partially account for this, and market reconciliation implicitly absorbs some correlation information. But the structural assumption of independence remains, and it is an approximation.

128 historical matches is a small dataset. By the standards of sports modeling, where practitioners typically work with thousands of league matches, 128 World Cup matches from 2018 and 2022 is a limited calibration sample. Calibration metrics and temperature estimates carry meaningful statistical uncertainty at this sample size. The model acknowledges this by applying modest shrinkage at every stage and treating single-match results with appropriate skepticism.

Lambda uncertainty is a fixed prior, not a per-match estimate. The +/-12% uncertainty used for confidence interval calculations is a conservative global assumption. The true uncertainty for a given match depends on the quality and recency of available data for each team. For teams with extensive, recent competitive history, 12% may overstate the uncertainty. For teams with limited data from distant qualifying regions, 12% may understate it.

Odds move between prediction and kickoff. Edge estimates are calculated at the time the pipeline runs. By the time you see them, the market may have moved. Always verify current odds

at your book before acting on any signal in the edge report. A +6% edge computed at 8:00 AM UTC may have narrowed or disappeared by the time the match kicks off.

Positive expected value does not guarantee positive returns in any finite sample. Even a well-calibrated model with genuine edge will experience losing streaks. The edge estimates are probabilistic – they describe long-run expectations, not individual outcomes. The Kelly sizing and confidence interval filter are designed to manage variance, not eliminate it.

The goal of this model is not to claim certainty about unpredictable events. It is to build the best available probability estimates from the available evidence, express them consistently, and be transparent about where those estimates are strong and where they are not. Football remains genuinely hard to predict. A good model makes you more informed, not omniscient.

All probabilities represent regulation time (90 minutes plus stoppage time) only. Extra time and penalties are excluded. This article is for informational and educational purposes. Please gamble responsibly.